

Nature Portfolio Reporting Summary

Completed responses prepared from the official Nature Portfolio fields. Last updated 19 May 2026.

Header

Corresponding author(s)	Provided in the editorial cover letter and submission metadata for double-anonymous peer review.
Last updated by author(s)	19 May 2026
Manuscript	Foundation-model morphology maps residual programs in spatial-omics contours.

Statistics

Exact sample sizes	The exact sample size for each association table is reported as the number of registered spatial-omics contours. Primary breast S3 residual associations use 157 paired H&E-WTA contours; cervical stress-test associations use 215 paired H&E-WTA contours. Source-data and Supplementary Data CSVs report the contour count for each ranked association and robustness analysis.
Data collection	No new experimental data were collected for this manuscript. The computational analysis uses all available paired contours passing the locked contour/WTA/image matching workflow from public/vendor 10x Genomics Atera WTA breast and cervical example datasets.
Statistical tests	Program-feature associations were tested with rank-residualized partial Spearman correlations. Spatial-null empirical P values are two-sided and computed from compartment-aware permutations. Robustness checks include block bootstrap, leave-one-block analysis, local mismatch controls, centroid jitter, registration perturbation and nested spatial holdout.
Covariates	Models include the spatial-omics-derived contour label where it is nonconstant, centroid-position covariates and boundary-distance summaries or bins. In breast S3 analyses the contour label is constant and is therefore omitted from the model matrix.
Assumptions	The analysis assumes that contour registration, tile embeddings and WTA program summaries are fixed outputs of the locked preprocessing workflow. Multiplicity is handled through ranked candidate reporting and robustness filters rather than by claiming cohort-level discovery.
Descriptive statistics	Where relevant, source-data files provide contour counts, residual association statistics, empirical P values and robustness summaries.

Null hypothesis	The null hypothesis is that no residual association remains between morphology embedding summaries and WTA program summaries after the specified covariates and spatial structure are accounted for.
Bayesian analysis	No Bayesian estimation was used.
Hierarchical analysis	No hierarchical random-effects model was used; spatial block and holdout analyses are sensitivity checks.
Effect sizes	Effect sizes are partial Spearman rho values, or the corresponding robustness-stage statistics, with empirical permutation P values and stability summaries reported in machine-readable source data.

Software and code

Data collection	No new instrument-control, wet-lab acquisition or participant-recruitment software was used by the authors. Public/vendor example datasets were downloaded and analyzed computationally.
Data analysis	Analyses were run with Python workflows in pyXenium/LazySlide scripts for mTM text embedding, direct WSI embedding, contour aggregation, statistical defense, figure composition and packaging. Exact public code and release identifiers are listed in the manuscript Code availability section and editorial cover letter. Source-data CSVs and logs in the submission package identify the outputs used for figures and robustness tables.

Data

Availability	Source data for Fig. 1 and supplementary figures are included in the submission package. Separate machine-readable Supplementary Data CSVs are included for robustness tables. Raw vendor inputs are public 10x Genomics Atera WTA and H&E example datasets and are not redistributed with this submission package; the manuscript Data availability statement provides the dataset URLs.
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Human participants, data and materials

Sex and gender	Sex and gender were not analyzed because the public/vendor example datasets do not provide participant-level attributes sufficient for inference in this computational secondary analysis.
Race and ethnicity	Race, ethnicity and related participant-level characteristics were not analyzed because the public/vendor example datasets do not provide participant-level attributes sufficient for inference in this computational secondary analysis.

Population characteristics	No participant-level clinical demographics, outcomes or identifiers were used. The study did not generate new identifiable participant data.
Recruitment	No participant recruitment was conducted by the authors; the work reuses public/vendor spatial WTA and H&E example data.
Ethics oversight	No new human sample collection, intervention or identifiable human-subject research was performed by the authors. The public/vendor datasets are cited as third-party example data; original consent and ethics oversight belong to the source data provider.

Field-specific reporting

Field-specific reporting	Life sciences.
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Life sciences study design

Sample size	All available paired contours passing the locked contour/WTa/image matching workflow were analyzed for the reported breast and cervical association sets.
Data exclusions	Contours without matched H&E-WTA summaries or required program/covariate values were excluded from the corresponding association table; analyzed contour counts are exposed in source-data files.
Replication	Cervical Atera WTA is treated as a second epithelial cancer stress test, not as a direct replication of breast luminal estrogen-response biology. PLIP and UNI provide model-sensitivity evidence at the program-family level. No IHC, protein-validation, clinical deployment or morphology-only diagnostic claim is made.
Randomization	No experimental randomization was introduced because this was a computational secondary analysis. Null distributions were generated by spatial/contour-stratified permutations and robustness resampling.
Blinding	No experimental group allocation or blinding was introduced. Analyses operate on locked preprocessing outputs and public/vendor datasets.

Reporting for specific materials, systems and methods

Antibodies	Antibodies were not involved because the study did not perform wet-lab immunostaining, antibody-based detection or antibody validation.
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Eukaryotic cell lines	Eukaryotic cell lines were not involved because the study used public/vendor tissue-section spatial WTA and H&E example datasets.
Palaeontology and archaeology	Palaeontological or archaeological specimens were not involved because the study concerns human cancer tissue example datasets.
Animals and field-collected samples	Animals, wild animals and field-collected biological samples were not involved because the study reanalyzes public/vendor human tissue spatial-omics example data.
Clinical data	Clinical trial or clinical-outcome data were not involved because the study does not use patient-level intervention, treatment, outcome or trial data.
Dual use research of concern	Dual-use research of concern was not involved because the work is computational morphology-spatial-omics analysis and does not alter pathogens, toxins, host range, transmissibility, virulence or resistance.
Plants	Plants or novel plant genotypes were not involved.
ChIP-seq	ChIP-seq was not involved because the study analyzes spatial WTA and H&E images rather than chromatin immunoprecipitation sequencing.
Flow cytometry	Flow cytometry was not involved because no cytometry acquisition, gating or cell-sorting experiment was performed.
MRI-based neuroimaging	MRI-based neuroimaging was not involved because no neuroimaging data or MRI analysis was used.